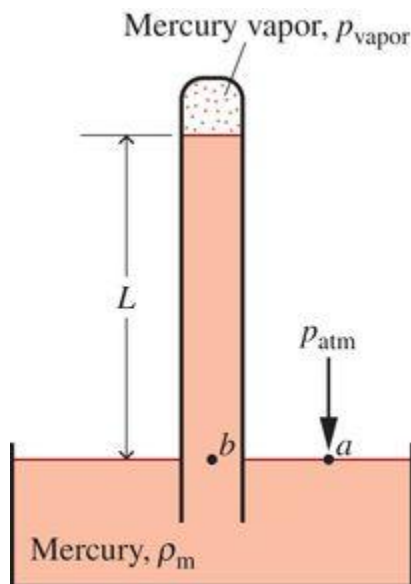


Homework 1 – Due January 27 at 11:59 PM

Please submit a pdf of completed HW through Canvas with all problems in a single file. You must show your work to earn full credit. Late work will not be accepted.

1. A closed vessel of volume 0.1 m^3 holds 3×10^{25} molecules of carbon dioxide.
 - a. Determine the number of moles present. (*Answer check: 49.82 mol*)
 - b. Determine the specific volume of carbon dioxide in the closed vessel. (*Answer check: $0.04560 \text{ m}^3/\text{kg}$*)
2. A mercury barometer measures pressure in the atmosphere using the hydrostatic pressure of a mercury column in a closed tube. If atmospheric pressure is 101.4 kPa , and the height of the mercury column L is 760 mm , what is the pressure of the mercury vapor at the top of the barometer? The density of mercury is 13550 kg/m^3 , and the acceleration due to gravity is 9.81 m/s^2 . (Refer to equation 1.6c from the textbook.) (*Answer check: 376.6 Pa*)



3. A mass of 10 kg undergoes a process during which there is a heat transfer to the mass at a rate of 4 kJ/kg , an elevation decrease of 30 m , and an increase in velocity from 20 m/s to 40 m/s . The specific internal energy decreases by 3 kJ/kg , and the acceleration due to gravity is 9.81 m/s^2 . Determine the work for the process (mind your units). (*Answer check: 66.94 kJ*)

4. A gas contained in a piston-cylinder assembly undergoes a process between end states 1 and 2, where $P_1 = 100 \text{ kPa}$, $V_1 = 1 \text{ m}^3$, $U_1 = 400 \text{ kJ}$, and $P_2 = 1000 \text{ kPa}$, $V_2 = 0.1 \text{ m}^3$, $U_2 = 500 \text{ kJ}$. Assume negligible changes in KE and PE.
- Assuming the process consists of a constant-volume process from state 1 to a pressure of 1000 kPa, followed by a constant pressure process to state 2, sketch the process on P-V coordinates, and evaluate the work and heat transfer. (*Answer check: $W_{12} = -900 \text{ kJ}$, $Q_{12} = -800 \text{ kJ}$*)
 - Assuming the pressure-volume relation is $PV = \text{constant}$, sketch the process on P-V coordinates, and evaluate the work and heat transfer. (*Answer check: $W_{12} = -230.3 \text{ kJ}$, $Q_{12} = -130.3 \text{ kJ}$*)
5. Air is contained in a vertical piston-cylinder assembly by a piston of mass 50 kg and having a face area of 0.01 m^2 . The mass of the air is 5 g, and the air initially occupies a volume of 5 liters. The atmosphere exerts a pressure of 100 kPa on the top of the piston. The volume of the air slowly decreases to 0.002 m^3 as the specific internal energy of the air decreases by 260 kJ/kg. Neglecting friction between the piston and cylinder wall, determine the heat transfer to the air. (*Answer check: $Q = -1.747 \text{ kJ}$*)